YAHOO STOCK

S&P 500

OBJECTIVES

- Analyze Trends
- Measure stock price volatility
- Develop forecasting models (ARIMA & Random Forest)
- Compare model performance

EXPLORATORY DATA ANALYSIS

DATA OVERVIEW

1,825 Records

7 Variables

23/11/2015 to 20/11/2020

(5 Years)

VARIABLES

- Date
- High
- Low
- Open
- Close
- Volume
- Adj Close

Dataset have 7 variables

DATA PREPARATION

Missing Values: None

Outliers: None

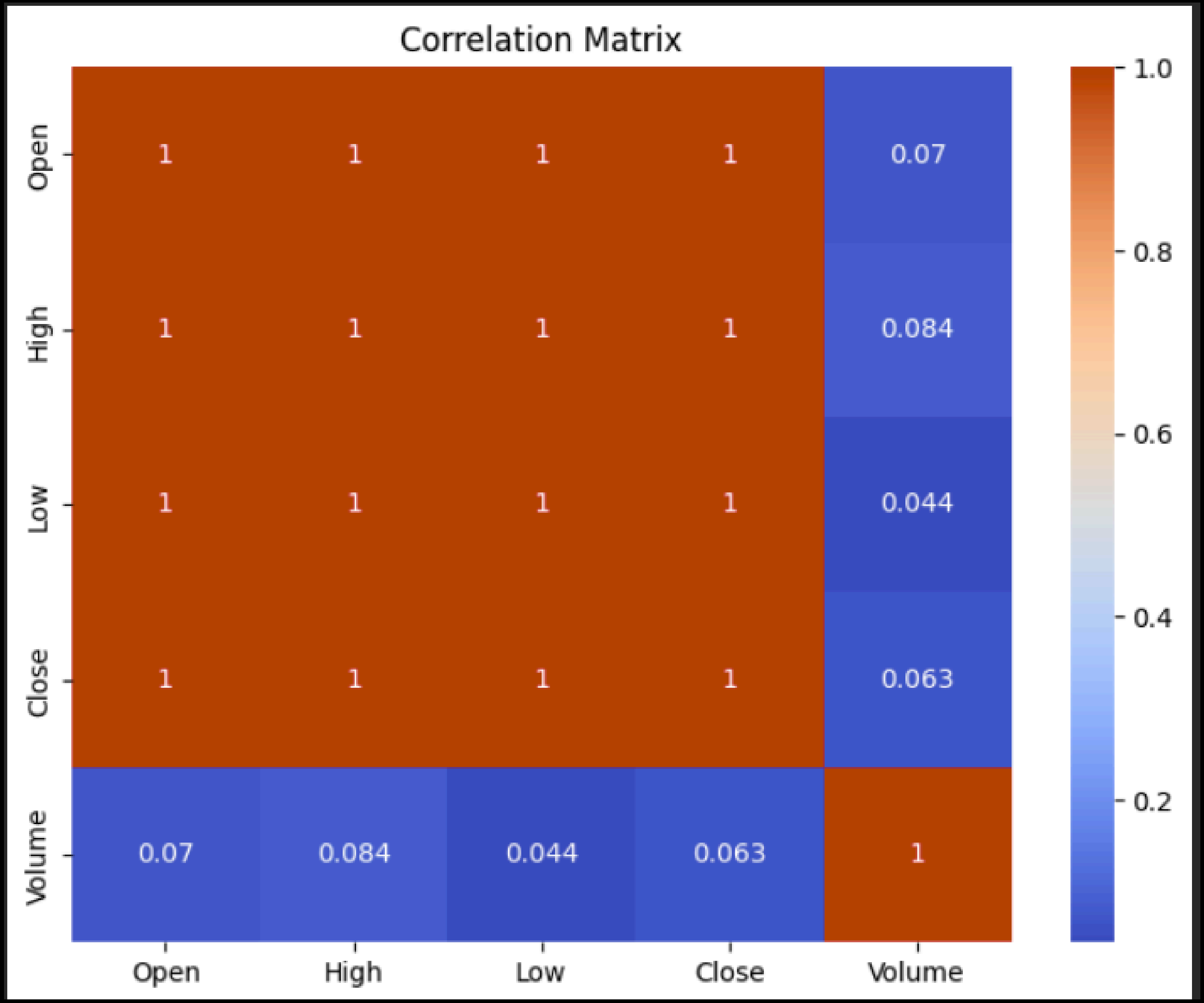
DESCRIPTIVE STATISTICS

CLOSE PRICE

MEAN	2647.856
MAX	3626.91
MIN	1829.079
STANDARD DEVIATION	407.3012

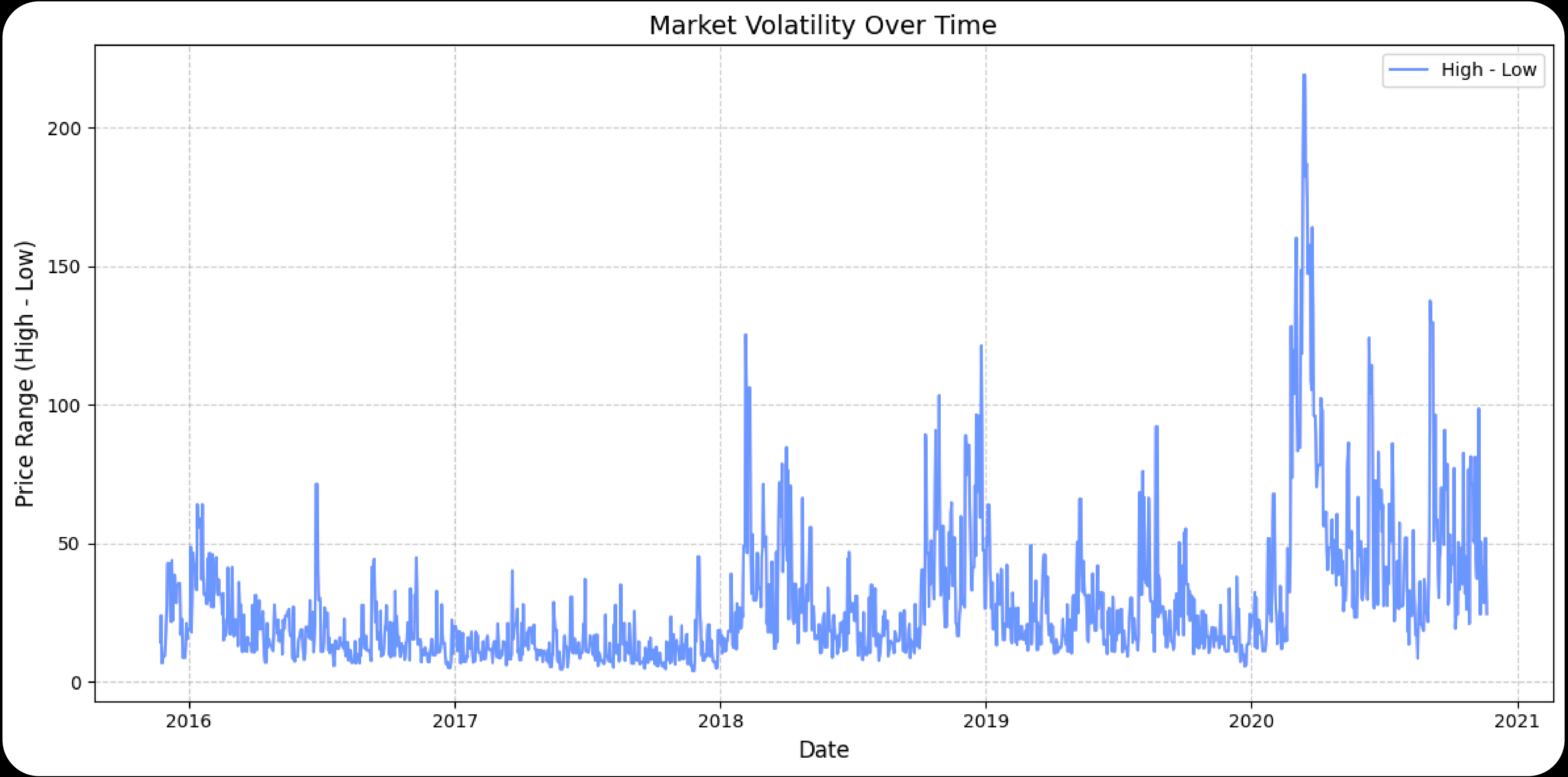
DISTRIBUTION OF CLOSING PRICES

CORRELATION MATRIX



CLOSE PRICE WITH TREND

VOLATILITY



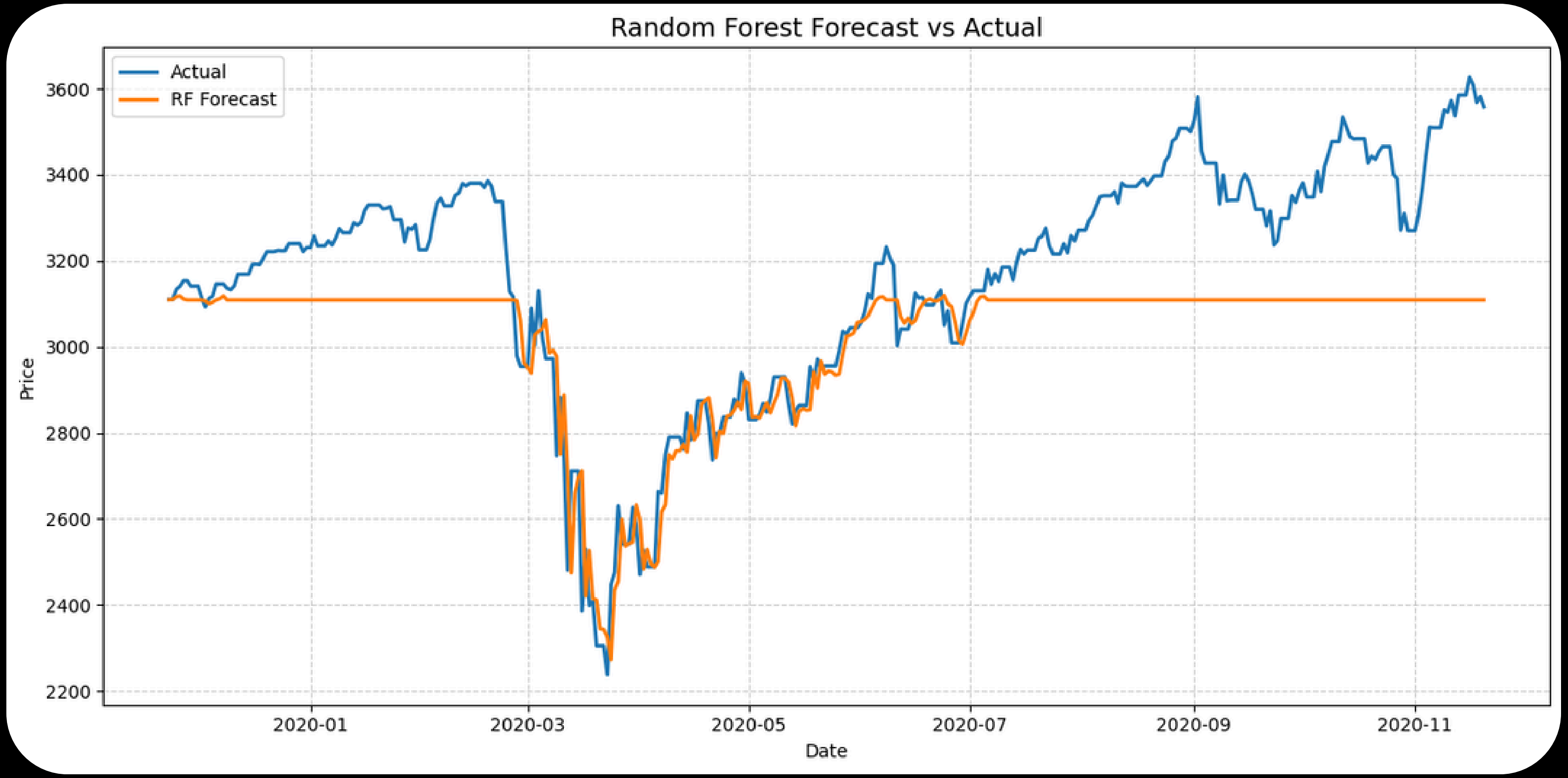
Market Volatility Over Time
High - Low
Price Range (High - Low)
0
50
100
150
200
2016
2017
2018
2019
2020
2021
Date

MODELING

MODELING

TIME SERIES
ARIMA

MACHINE LEARNING
RANDOM FOREST



RESULT

ARIMA		RANDOM FOREST	
MAE	228.095	MAE	156.195
RMSE	278.605	RMSE	201.992

CONCLUSION

"The stock market is often affected by unpredictable events, like Black Swan events, which limit the accuracy of our models."

"ARIMA is good for linear data with a clear trend. However, stock data is non-linear and complex. This is why Random Forest performs better, as it can learn these complex patterns."

THANK YOU

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